

(KTXFI2EBNF) Physics II. Test 2 (Midterm Retake)  
December 8, 2025

1. An electron has a kinetic energy of 0.5 keV.

(a) What is its velocity?

Convert to SI:  $K = 0.5 \text{ keV} = 500 \text{ eV} = 500 \cdot (1.602176634 \cdot 10^{-19}) \text{ J} \approx 8.01088317 \cdot 10^{-17} \text{ J}$ .  
Using the nonrelativistic kinetic energy formula

$$K = \frac{1}{2}mv^2,$$

with  $m = m_e = 9.1 \cdot 10^{-31} \text{ kg}$  gives

$$v = \sqrt{\frac{2K}{m}} \approx \sqrt{\frac{2 \cdot 8.01088317 \cdot 10^{-17}}{9.1 \cdot 10^{-31}}} \approx 1.3269 \cdot 10^7 \text{ m/s}.$$

(b) What is its momentum?

$$p = mv \approx (9.1 \cdot 10^{-31} \text{ kg}) \cdot (1.3269 \cdot 10^7 \text{ m/s}) \approx 1.2075 \cdot 10^{-23} \text{ kg m/s}.$$

(c) What is the wavelength of the associated de Broglie wave?

Using  $\lambda = h/p$  with  $h = 6.62607015 \cdot 10^{-34} \text{ J}\cdot\text{s}$ ,

$$\lambda = \frac{h}{p} \approx \frac{6.62607015 \cdot 10^{-34}}{1.2075 \cdot 10^{-23}} \approx 5.4876 \cdot 10^{-11} \text{ m} \approx 0.0549 \text{ nm}.$$

(10 points)

2. A metal surface is illuminated with light of wavelength  $9 \cdot 10^{-8} \text{ m}$ . What is the velocity of the emitted electrons if the photoelectric effect starts at a wavelength of  $2.88 \cdot 10^{-7} \text{ m}$ ? (Use  $m_e = 9.1 \cdot 10^{-31} \text{ kg}$ .)  
Photon energy:  $E_{\text{photon}} = hc/\lambda$ . The electron kinetic energy equals the excess energy:

$$K = hc \left( \frac{1}{\lambda_{\text{in}}} - \frac{1}{\lambda_{\text{th}}} \right),$$

with  $h = 6.62607015 \cdot 10^{-34} \text{ J}\cdot\text{s}$  and  $c = 2.99792458 \cdot 10^8 \text{ m/s}$ . Substituting  $\lambda_{\text{in}} = 9.0 \cdot 10^{-8} \text{ m}$  and  $\lambda_{\text{th}} = 2.88 \cdot 10^{-7} \text{ m}$  gives

$$K \approx 1.5174 \cdot 10^{-18} \text{ J}.$$

Then the emitted-electron velocity (nonrelativistic approximation)

$$v = \sqrt{\frac{2K}{m_e}} \approx \sqrt{\frac{2 \cdot 1.5174 \cdot 10^{-18}}{9.1 \cdot 10^{-31}}} \approx 1.8262 \cdot 10^6 \text{ m/s}.$$

(10 points)

3. In a Compton scattering experiment, X-rays with a wavelength of  $0.2 \text{ nm}$  are used.  
( $m_e = 9.1 \cdot 10^{-31} \text{ kg}$ )

(a) At what scattering angle does the wavelength of the radiation increase by 1%?

Initial wavelength  $\lambda_0 = 0.2 \text{ nm} = 2.00 \cdot 10^{-10} \text{ m}$ . A 1% increase means  $\Delta\lambda = 0.01\lambda_0 = 2.00 \cdot 10^{-12} \text{ m}$ . The Compton wavelength is

$$\lambda_C = \frac{h}{m_e c} \approx \frac{6.62607015 \cdot 10^{-34}}{(9.1 \cdot 10^{-31})(2.99792458 \cdot 10^8)} \approx 2.4288 \cdot 10^{-12} \text{ m}.$$

Compton formula:

$$\Delta\lambda = \lambda_C(1 - \cos\theta).$$

Hence

$$1 - \cos\theta = \frac{\Delta\lambda}{\lambda_C} \approx \frac{2.00 \cdot 10^{-12}}{2.4288 \cdot 10^{-12}} \approx 0.82345,$$

so

$$\cos\theta \approx 0.17655, \quad \theta \approx \arccos(0.17655) \approx 79.83^\circ.$$

(b) At what angle does the wavelength become 0.01

Now  $\Delta\lambda = 0.0001\lambda_0 = 2.00 \cdot 10^{-14} \text{ m}$ , therefore

$$1 - \cos\theta = \frac{2.00 \cdot 10^{-14}}{2.4288 \cdot 10^{-12}} \approx 0.0082378,$$

so

$$\cos\theta \approx 0.9917622, \quad \theta \approx \arccos(0.9917622) \approx 7.36^\circ.$$

(10 points)

4. A light source has a power output of 20 W.

(a) What is the frequency of the light if the entire energy of the source is emitted as light with wavelength  $\lambda = 800 \text{ nm}$ ?

$$f = \frac{c}{\lambda} = \frac{2.99792458 \cdot 10^8}{800 \cdot 10^{-9}} \text{ Hz} \approx 3.7474 \cdot 10^{14} \text{ Hz}.$$

(b) How many photons are emitted by the light source per second?

Each photon has energy  $E = hf$ , therefore the photon emission rate is

$$\dot{N} = \frac{P}{hf} = \frac{P\lambda}{hc}.$$

With  $P = 20 \text{ W}$  and  $\lambda = 800 \cdot 10^{-9} \text{ m}$ ,

$$\dot{N} \approx \frac{20 \cdot 800 \cdot 10^{-9}}{(6.62607015 \cdot 10^{-34})(2.99792458 \cdot 10^8)} \approx 8.0546 \cdot 10^{19} \text{ photons/s}.$$

(10 points)

5. We study an electron in a radioactive particle beam as it passes through the laboratory. It is found that, as measured in the laboratory, the average lifetime of the particle is 60 ns. The average lifetime measured in the particle rest frame is 5 ns. What is the propagation velocity of the particle beam? Determine the value of  $\frac{m}{m_0}$  for the electron! Calculate the rest energy of an electron corresponding to its rest mass ( $m_e = 9.1 \cdot 10^{-31} \text{ kg}$ ) in joules and in electronvolts.

Time dilation gives  $\tau = \gamma\tau_0$ , hence

$$\gamma = \frac{\tau}{\tau_0} = \frac{60 \text{ ns}}{5 \text{ ns}} = 12.$$

Therefore

$$\beta = \frac{v}{c} = \sqrt{1 - \frac{1}{\gamma^2}} = \sqrt{1 - \frac{1}{144}} \approx 0.99652,$$

and

$$v \approx 0.99652c \approx 2.9875 \cdot 10^8 \text{ m/s}.$$

The relativistic mass factor is

$$\frac{m}{m_0} = \gamma \approx 12.$$

The rest energy of the electron is

$$E_0 = m_e c^2 \approx (9.1 \cdot 10^{-31} \text{ kg}) \cdot (2.99792458 \cdot 10^8 \text{ m/s})^2 \approx 8.1787 \cdot 10^{-14} \text{ J}.$$

In electronvolts,

$$E_0 \approx \frac{8.1787 \cdot 10^{-14} \text{ J}}{1.602176634 \cdot 10^{-19} \text{ J/eV}} \approx 5.1047 \cdot 10^5 \text{ eV} \approx 510.5 \text{ keV} \approx 0.5105 \text{ MeV}.$$

(10 points)

Requirements for obtaining a signature: achieving at least 40

Grades: 0–25 points: Fail (1), 26–30 points: Pass (2), 31–37 points: Satisfactory (3), 38–45 points: Good (4), 46–50 points: Excellent (5).

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